SAMIAA LEARNING | RAG CHUNKING LAB

Printable companion matching the Chunking Lab UI.
Source document (left) * Chunks (right) * Same layout as the app.

How to read this guide

LEFT - Source document with color-coded highlights. Each color = one chunk.

RIGHT - Chunk cards with ID, labels, embedded status, char positions.

Parent-child pages show Parents (not embedded) and Children (embedded) separately.

Open the app: `cd frontend && npm run dev` -> http://localhost:5173

Strategies covered

- **Fixed-size** - Split every N characters (breaks on word boundaries).
- **Overlap** - Sliding window with shared text between adjacent chunks.
- **Recursive** - Prefer headings → paragraphs → sentences → characters.
- **Semantic** - Split when sentence similarity drops (TF-IDF vectors in-browser).
- **Parent–child** - Section = parent (stored, not embedded). Sentences = children (embedded).
- **Proposition** - One atomic fact per sentence, rewritten to stand alone.

1. Recursive - Employee Handbook

Document: Employee Handbook **Strategy:** Recursive
Structured policy doc with headings — best for recursive and parent–child chunking.
Prefer headings → paragraphs → sentences → characters.
Chunk size: 280 chars
1 total chunks * 1 searchable (embedded)

Source document

```
## Remote Work Policy
Employees may work remotely up to three days per week. Managers must approve schedules by Friday.
Equipment stipends are $500 per year. Security training is mandatory before VPN access is
granted.
```

Chunks

ALL CHUNKS
| recursive_1 ## Remote Work Policy * heading [embedded]
Remote Work Policy Employees may work remotely up to three days per week. Managers must approve schedules by Friday. Equipment stipends are $500 per year. Security training is mandatory before VPN access is granted.
chars 0-218 * 218 chars

2. Fixed-size - Employee Handbook (compare cuts)

Document: Employee Handbook **Strategy:** Fixed-size
Structured policy doc with headings — best for recursive and parent–child chunking.
Split every N characters (breaks on word boundaries).
Chunk size: 280 chars - notice mid-section cuts vs recursive
1 total chunks * 1 searchable (embedded)

Source document

```
## Remote Work Policy
Employees may work remotely up to three days per week. Managers must approve schedules by Friday.
Equipment stipends are $500 per year. Security training is mandatory before VPN access is
granted.
```

Chunks

ALL CHUNKS
| fixed_1 character [embedded]
Remote Work Policy Employees may work remotely up to three days per week. Managers must approve schedules by Friday. Equipment stipends are $500 per year. Security training is mandatory before VPN access is granted.
chars 0-218 * 218 chars

3. Parent-child - Employee Handbook

Document: Employee Handbook **Strategy:** Parent–child
Structured policy doc with headings — best for recursive and parent–child chunking.
Section = parent (stored, not embedded). Sentences = children (embedded).
5 total chunks * 4 searchable (embedded) * 1 parents (text only)

Source document

```
## Remote Work Policy
Employees may work remotely up to three days per week. Managers must approve schedules by Friday.
Equipment stipends are $500 per year. Security training is mandatory before VPN access is
granted.
```

Chunks

PARENTS (stored - not embedded)
| parent_1 ## Remote Work Policy * heading [not embedded]
Remote Work Policy Employees may work remotely up to three days per week. Managers must approve schedules by Friday. Equipment stipends are $500 per year. Security training is mandatory before VPN access is granted.
chars 0-218 * 218 chars

CHILDREN (embedded for search)
| child_1 → parent_1 * sentence [embedded]
Employees may work remotely up to three days per week.
chars 22-77 * 54 chars * parent: parent_1

| child_2 → parent_1 * sentence [embedded]
Managers must approve schedules by Friday.
chars 77-120 * 42 chars * parent: parent_1

| child_3 → parent_1 * sentence [embedded]
Equipment stipends are $500 per year.
chars 120-158 * 37 chars * parent: parent_1

| child_4 → parent_1 * sentence [embedded]
Security training is mandatory before VPN access is granted.
chars 158-218 * 60 chars * parent: parent_1

4. Semantic - Employee Handbook

Document: Employee Handbook **Strategy:** Semantic
Structured policy doc with headings — best for recursive and parent–child chunking.
Split when sentence similarity drops (TF-IDF vectors in-browser).
Similarity threshold: 0.25
4 total chunks * 4 searchable (embedded)

Source document

```
## Remote Work Policy
Employees may work remotely up to three days per week. Managers must approve schedules by Friday.
Equipment stipends are $500 per year. Security training is mandatory before VPN access is
granted.
```

Chunks

ALL CHUNKS

| **semantic_1** topic start * sentence [embedded]
Remote Work Policy Employees may work remotely up to three days per week.
chars 0-77 * 76 chars

| **semantic_2** topic break * sim 0.00 < 0.25 * sentence [embedded]
Managers must approve schedules by Friday.
chars 77-120 * 42 chars * sim to prev: 0.00

| **semantic_3** topic break * sim 0.00 < 0.25 * sentence [embedded]
Equipment stipends are $500 per year.
chars 120-158 * 37 chars * sim to prev: 0.00

| **semantic_4** topic break * sim 0.00 < 0.25 * sentence [embedded]
Security training is mandatory before VPN access is granted.
chars 158-218 * 60 chars * sim to prev: 0.00

5. Proposition - Benefits Policy

Document: Benefits Policy **Strategy:** Proposition
Dense paragraph with multiple facts — best for proposition (atomic) chunking.
One atomic fact per sentence, rewritten to stand alone.
3 total chunks * 3 searchable (embedded)

Source document

```
The Acme Plan covers dental after a six-month waiting period and vision after ninety days for full-time staff. Part-time employees are eligible for vision only after one hundred eighty days. Dental coverage is not available to part-time employees.
```

Chunks

ALL CHUNKS

| **prop_1** atomic fact 1 * sentence [embedded]
Benefits Policy: the Acme Plan covers dental after a six-month waiting period and vision after ninety days for full-time staff.
chars 0-111 * 127 chars

| **prop_2** atomic fact 2 * sentence [embedded]
Benefits Policy: part-time employees are eligible for vision only after one hundred eighty days.
chars 111-191 * 96 chars

| **prop_3** atomic fact 3 * sentence [embedded]
Benefits Policy: dental coverage is not available to part-time employees.
chars 191-247 * 73 chars

6. Fixed-size - Benefits Policy (compare)

Document: Benefits Policy **Strategy:** Fixed-size
Dense paragraph with multiple facts — best for proposition (atomic) chunking.
Split every N characters (breaks on word boundaries).
Chunk size: 200 - multiple facts in one chunk
2 total chunks * 2 searchable (embedded)

Source document

```
The Acme Plan covers dental after a six-month waiting period and vision after ninety days for
full-time staff. Part-time employees are eligible for vision only after one hundred eighty days.
Dental coverage is not available to part-time employees.
```

Chunks

ALL CHUNKS

| fixed_1 character [embedded]
The Acme Plan covers dental after a six-month waiting period and vision after ninety days for full-time staff. Part-time employees are eligible for vision only after one hundred eighty days. Dental
chars 0-197 * 197 chars

| fixed_2 character [embedded]
coverage is not available to part-time employees.
chars 197-247 * 49 chars

7. Semantic - Company Update (3 topics)

Document: Company Update (mixed topics) **Strategy:** Semantic
One article with several topics and no clear headings — compare fixed-size vs semantic.
Split when sentence similarity drops (TF-IDF vectors in-browser).
Similarity threshold: 0.25 - launch / pricing / hiring
8 total chunks * 8 searchable (embedded)

Source document

```
Internal Company Update — March 2026

We shipped version 2.0 last week with offline sync and faster search. Customers have been asking for offline mode for over a year, and early feedback has been very positive.

Starting May 1, the Pro plan will be $29 per month instead of $39. Existing customers keep their current price until their contract renews. The Team plan pricing is unchanged at $99 per month.

We are hiring two backend engineers and one designer for the platform team. Applications close on April 15. Referrals receive a $2,000 bonus after the new hire completes 90 days.
```

Chunks

ALL CHUNKS

semantic_1 topic start * sentence [embedded]
0 last week with offline sync and faster search.
chars 59-108 * 48 chars

semantic_2 topic break * sim 0.09 < 0.25 * sentence [embedded]
Customers have been asking for offline mode for over a year, and early feedback has been very positive.
chars 108-213 * 103 chars * sim to prev: 0.09

semantic_3 topic break * sim 0.00 < 0.25 * sentence [embedded]
Starting May 1, the Pro plan will be $29 per month instead of $39.
chars 213-280 * 66 chars * sim to prev: 0.00

semantic_4 topic break * sim 0.00 < 0.25 * sentence [embedded]
Existing customers keep their current price until their contract renews.
chars 280-353 * 72 chars * sim to prev: 0.00

semantic_5 topic break * sim 0.00 < 0.25 * sentence [embedded]
The Team plan pricing is unchanged at $99 per month.
chars 353-407 * 52 chars * sim to prev: 0.00

semantic_6 topic break * sim 0.12 < 0.25 * sentence [embedded]
We are hiring two backend engineers and one designer for the platform team.
chars 407-483 * 75 chars * sim to prev: 0.12

semantic_7 topic break * sim 0.00 < 0.25 * sentence [embedded]
Applications close on April 15.
chars 483-515 * 31 chars * sim to prev: 0.00

semantic_8 topic break * sim 0.00 < 0.25 * sentence [embedded]
Referrals receive a $2,000 bonus after the new hire completes 90 days.
chars 515-585 * 70 chars * sim to prev: 0.00

8. Fixed-size - Company Update (compare)

Document: Company Update (mixed topics) **Strategy:** Fixed-size
One article with several topics and no clear headings — compare fixed-size vs semantic.
Split every N characters (breaks on word boundaries).
Chunk size: 280 - topics may mix in one chunk
3 total chunks * 3 searchable (embedded)

Source document

```
Internal Company Update — March 2026

We shipped version 2.0 last week with offline sync and faster search. Customers have been asking for offline mode for over a year, and early feedback has been very positive.

Starting May 1, the Pro plan will be $29 per month instead of $39. Existing customers keep their current price until their contract renews. The Team plan pricing is unchanged at $99 per month.

We are hiring two backend engineers and one designer for the platform team. Applications close on April 15. Referrals receive a $2,000 bonus after the new hire completes 90 days.
```

Chunks

ALL CHUNKS

| **fixed_1** character [embedded]
Internal Company Update — March 2026 We shipped version 2.0 last week with offline sync and faster search. Customers have been asking for offline mode for over a year, and early feedback has been very positive. Starting May 1, the Pro plan will be $29 per month instead of $39.
chars 0-279 * 279 chars

| **fixed_2** character [embedded]
Existing customers keep their current price until their contract renews. The Team plan pricing is unchanged at $99 per month. We are hiring two backend engineers and one designer for the platform team. Applications close on April 15. Referrals receive a $2,000 bonus after the
chars 279-557 * 277 chars

| **fixed_3** character [embedded]
new hire completes 90 days.
chars 557-585 * 27 chars

9. Overlap - Security FAQ

Document: Security FAQ (flat text) **Strategy:** Overlap
Flat Q&A without headings — good for fixed-size and overlap demos.
Sliding window with shared text between adjacent chunks.
*Chunk size: 200 * Overlap: 15%*
2 total chunks * 2 searchable (embedded)

Source document

```
Q: Do I need VPN for remote work?
A: Yes. VPN is required for all remote access to internal systems. Security training must be
completed before VPN credentials are issued.

Q: How often should I.

Q: How often should I change my password?
A: Passwords must be rotated every 90 days. Use at least 12 characters with mixed case, numbers,
and symbols.
```

Chunks

ALL CHUNKS

| overlap_1 overlap 30 chars with previous * character [embedded]
Q: Do I need VPN for remote work? A: Yes. VPN is required for all remote access to internal systems. Security training must be completed before VPN credentials are issued. Q: How often should I
chars 0-194 * 194 chars

| overlap_2 overlap 30 chars with previous * character [embedded]
. Q: How often should I change my password? A: Passwords must be rotated every 90 days. Use at least 12 characters with mixed case, numbers, and symbols.
chars 170-324 * 154 chars

Parent-child: what gets embedded?

Piece	Embedded?	Stored where	Used for
Child (sentence)	Yes	Vector DB (later)	Search
Parent (section)	No	DB / doc store	LLM context

Retrieval flow: Query -> search children -> read parent_id -> fetch parent text -> send to LLM
Mental model: children = search snippets * parent = full page you open after the match

Strategy comparison

Strategy	Precision	Context	Best fit
Fixed-size	Medium	Variable	Prototypes, flat text
Overlap	Higher	Better	Narrative, policies
Recursive	High	High	Structured docs
Semantic	High	High	Mixed topics
Parent-child	Very high	Very high	Long manuals
Proposition	Very high	Focused	Factoid Q&A